

Non-Circular G_F Chain Framework

BVP-Gated Derivation Structure

EDC Project

2026-01-29

Epistemic Status

Skeleton structure: [Der] — dimensional analysis and chain logic

Parameter mappings: [Dc] — conditional on mode profiles from BVP

Numerical values: [OPEN] — blocked by BVP solution (OPR-21)

Goal: Define a chain from 5D action $\rightarrow G_F$ with NO circular v input.

1 Step 0: Circularity Audit

1.1 The Circular Dependency

The Standard Model relation:

$$v = (\sqrt{2}G_F)^{-1/2} = 246.22 \text{ GeV} \quad [\text{BL}] \quad (1)$$

defines the Higgs VEV from the measured G_F . Therefore:

Circularity Caveat

Using v to derive $M_W = gv/2$, then M_W to derive $G_F = g^2/(4\sqrt{2}M_W^2)$ is **circular**—it returns the input G_F by construction.

1.2 Where Circularity Occurs (File References)

File	Line(s)	Circular Element
11_gf_derivation.tex	119–127	$G_F = g^2/(4\sqrt{2}M_W^2)$ via v
docs/GF_CONSTRAINT_NOTE.md	B.2–B.3	v defined from G_F
docs/STATUS.md	38	G_F marked “circular”
docs/TODO.md	17	“Derive G_F without circular input”

1.3 What IS Independent

- $\sin^2 \theta_W = 1/4$ from $\mathbb{Z}_6$ subgroup counting [Der]
- $\alpha = 1/137.036$ [BL] (measured independently of G_F)

- $m_e = 0.511 \text{ MeV [BL]}$ (measured independently)
- $\delta = \hbar/(2m_p c) = 0.105 \text{ fm [Dc]}$

2 Step 1: Non-Circular Chain Definitions

2.1 The Forward Chain (No v Allowed)

Non-Circular Derivation Chain

$$\boxed{5\text{D Action} \xrightarrow{\text{norm}} g_5 \xrightarrow{\text{KK}} M_{\text{eff}} \xrightarrow{\text{BVP}} \{w_L, w_R, w_\phi\} \xrightarrow{\text{overlap}} I_4 \xrightarrow{\text{assemble}} G_F^{\text{EDC}}} \quad (2)$$

2.2 Definitions

5D Gauge Coupling g_5 : From canonical normalization of 5D gauge action:

$$S_{5D} = \int d^5x \sqrt{-g} \left(-\frac{1}{4g_5^2} F_{MN} F^{MN} \right) \quad (3)$$

Dimension: $[g_5^2] = [\text{length}] = [\text{energy}]^{-1} [\text{P}]$

Effective Mediator Mass M_{eff} : From KK reduction along extra dimension:

$$M_{\text{eff}}^2 = \lambda_0 \cdot \frac{1}{\delta^2} \quad (4)$$

where λ_0 is the lowest massive KK eigenvalue (dimensionless). [OPEN]

Mode Profiles $w_L(\chi), w_R(\chi), w_\phi(\chi)$: Solutions to the thick-brane Dirac/KG equations with asymmetric mass profile. Normalization: $\int d\chi |w|^2 = 1$. [OPEN]

Overlap Integral I_4 :

$$I_4 := \int d\chi w_L(\chi)^2 w_R(\chi)^2 w_\phi(\chi)^2 \quad (5)$$

Dimension: $[I_4] = [\text{length}]^{-1} = [\text{energy}]$. [OPEN]

Effective 4D Coupling g_{eff} :

$$g_{\text{eff}}^2 = g_5^2 \cdot I_g \quad (6)$$

where $I_g = \int d\chi w_\phi^2$ is the gauge boson overlap. [Dc]

Final Assembly:

$$\boxed{G_F^{\text{EDC}} = \frac{g_{\text{eff}}^2}{4\sqrt{2}M_{\text{eff}}^2}} \quad (7)$$

3 Step 2A: Derivable Skeleton (No BVP Needed)

3.1 Dimensional Skeleton [Der]

Using dimensional analysis alone:

$$[G_F] = [\text{energy}]^{-2} \quad (8)$$

$$[g_5^2] = [\text{length}] = [\text{energy}]^{-1} \quad (9)$$

$$[I_4] = [\text{energy}] \quad (10)$$

$$[M_{\text{eff}}^2] = [\text{energy}]^2 \quad (11)$$

Therefore the **unique dimensionally correct combination** is:

$$G_F^{\text{EDC}} = C \cdot \frac{g_5^2 \cdot I_4}{M_{\text{eff}}^2} \quad (12)$$

where C is a numerical factor from conventions ($C = 1/4\sqrt{2}$ for SM normalization).

3.2 Dimensionless Form [Der]

Define the dimensionless target:

$$X := G_F m_e^2 = 3.04 \times 10^{-12} \quad [\text{BL}] \quad (13)$$

Then the EDC prediction is:

$$X_{\text{EDC}} = C \cdot \frac{g_5^2 \cdot I_4 \cdot m_e^2}{M_{\text{eff}}^2} \quad (14)$$

3.3 Reduction Using EDC Scales [Dc]

If $M_{\text{eff}} \sim 1/\delta$ and $g_5^2 \sim \delta$ (natural 5D scaling):

$$X_{\text{EDC}} \sim C \cdot \delta \cdot I_4 \cdot m_e^2 \cdot \delta^2 = C \cdot I_4 \cdot m_e^2 \cdot \delta^3 \quad (15)$$

With $\delta = 0.105 \text{ fm} = 5.3 \times 10^{-4} \text{ GeV}^{-1}$:

$$X_{\text{EDC}} \sim C \cdot I_4 \cdot (0.511 \text{ MeV})^2 \cdot (5.3 \times 10^{-4} \text{ GeV}^{-1})^3 \quad (16)$$

Key insight: The δ^3 factor provides enormous suppression, potentially explaining the smallness of G_F .

4 Step 2B: BVP-Dependent Layer (Falsification Gates)

4.1 What BVP Must Output

Quantity	Definition	Status
$w_L(\chi)$	Left-handed fermion mode	[OPEN]
$w_R(\chi)$	Right-handed fermion mode	[OPEN]
$w_\phi(\chi)$	Mediator (scalar/gauge) mode	[OPEN]
λ_0	Lowest massive KK eigenvalue	[OPEN]
I_4	Overlap integral (5)	[OPEN]

4.2 Falsification Gates

Falsification Criteria (P3-3)

FAIL conditions:

1. **Overlap mismatch:** I_4 from BVP differs from constraint window by $> 10\times$

$$\text{FAIL if } I_4^{\text{BVP}} \notin [0.1, 10] \times I_4^{\text{required}} \quad (17)$$

2. **Mass inconsistency:** M_{eff} violates brane-thickness scaling

$$\text{FAIL if } M_{\text{eff}} \notin [0.1, 10] \times (1/\delta) \quad (18)$$

3. **Coupling incompatibility:** g_5 from action contradicts $\sin^2 \theta_W = 1/4$

$$\text{FAIL if } g_{\text{eff}}^2 \neq (4\pi\alpha/\sin^2 \theta_W) \times f(\text{overlaps}) \quad (19)$$

with $f \sim O(1)$.

4.3 Success Criteria

$$\text{SUCCESS: } \left| \frac{X_{\text{EDC}} - X_{\text{target}}}{X_{\text{target}}} \right| < 0.1 \quad (\pm 10\%) \quad (20)$$

5 Step 3: Toy Model Bounds (Feasibility Window)

5.1 Simplified Ansatz

Assume exponential mode profiles with width σ :

$$w_L(\chi) = \sqrt{\frac{1}{\sigma}} e^{-\chi/(2\sigma)} \quad (21)$$

$$w_R(\chi) = \sqrt{\frac{1}{\sigma}} e^{-(\chi-d)/(2\sigma)} \quad (\text{displaced by } d) \quad (22)$$

$$w_\phi(\chi) = \sqrt{\frac{1}{\delta}} e^{-\chi/(2\delta)} \quad (23)$$

5.2 Chirality Suppression Factor

The left-right overlap:

$$\varepsilon := \int d\chi w_L w_R = e^{-d/(2\sigma)} \quad (24)$$

For $d \gg \sigma$: $\varepsilon \ll 1$ (chirality separation).

5.3 Overlap Integral Estimate

For the four-point overlap:

$$I_4 \sim \frac{\varepsilon^2}{\sigma} \quad (\text{rough estimate}) \quad (25)$$

5.4 Required ε for Target Window

From (14), matching $X_{\text{target}} = 3 \times 10^{-12}$ requires:

$$I_4^{\text{required}} \sim \frac{X_{\text{target}}}{C \cdot g_5^2 \cdot m_e^2 / M_{\text{eff}}^2} \quad (26)$$

With $M_{\text{eff}} \sim 1/\delta \sim 1.9 \text{ GeV}$ and $g_5^2 \sim 4\pi\alpha\delta$:

$$I_4^{\text{required}} \sim 4 \times 10^{-5} \text{ GeV}^2 \sim (6 \text{ MeV})^2 \quad (27)$$

Feasibility: This scale is consistent with fermion localization at $\sim r_e^{-1}$.

5.5 Toy Window Summary

Parameter	Range for $X_{\text{EDC}} \approx X_{\text{target}}$	Status
ε (chirality suppression)	$10^{-3} - 10^{-2}$	Plausible
σ/δ (localization ratio)	$0.01 - 0.1$	Plausible
M_{eff}	$1 - 10 \text{ GeV}$	Plausible

Toy Model Caveat

This is a **feasibility window**, not an EDC derivation. The actual values must come from the thick-brane BVP solution (OPR-21).

6 Summary

P3-3 Framework Summary

Non-circular formula:

$$X_{\text{EDC}} = \frac{g_5^2 \cdot I_4 \cdot m_e^2}{4\sqrt{2} M_{\text{eff}}^2} \quad (28)$$

What is derived:

- Dimensional skeleton [Der]
- Independence from v [Der]
- $\sin^2 \theta_W = 1/4$ (separate prediction) [Der]

What is conditional [Dc]:

- g_5 from 5D action normalization (OPR-19)
- M_{eff} from KK eigenvalue (OPR-20)

What is OPEN (BVP-gated):

- Mode profiles w_L, w_R, w_ϕ (OPR-21)
- Overlap integral I_4
- Numerical G_F^{EDC} value

6.1 Epistemic Assessment

Component	Status	Blocking Issue
Framework existence	[Der] (GREEN)	—
Dimensional skeleton	[Der] (GREEN)	—
Circularity removed	[Der] (GREEN)	—
Toy feasibility	[I] (YELLOW)	—
g_5 value	[Dc]	OPR-19
M_{eff} value	[Dc]	OPR-20
I_4 value	[OPEN]	OPR-21 (BVP)
Numerical G_F	[OPEN]	OPR-21 (BVP)